HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed high-performance Kdb+/Q data pipelines for real-time market data ingestion and historical analysis, processing terabytes of data daily at Citadel Securities.
- Designed and implemented scalable Kdb+/Q solutions to support quantitative research and trading workflows, reducing query latency for complex analytical tasks by **30%**.
- Collaborated directly with quantitative researchers and traders to gather requirements and deliver technology solutions that directly impacted trading strategy execution and data analytics capabilities.
- Contributed to the architecture and evolution of a large-scale Kdb+/Q platform, enhancing its capacity for time-series analytics and distributed computing challenges across global markets.
- Wrote **Python** scripts to automate data validation and pre-processing for Kdb+/Q data feeds, improving data quality and reducing manual effort by an estimated **15 hours per week**.
- Managed production Kdb+/Q environments, ensuring high availability and performance for critical trading applications through proactive monitoring and incident response.
- Containerized Kdb+/Q services using **Docker** and orchestrated deployments on **AWS** infrastructure, enabling faster rollouts and improved resource management.

SKILLS

Core Technologies – Kdb+, Q, Python, Java, C++, SQL, Linux

Platforms & Systems – Time-Series Databases, Market Data Platforms, Distributed Systems, Global Trading Applications

Development Tools – Git, Jenkins, Docker, AWS, CloudWatch

Methodologies – Agile, Scrum, Test-Driven Development (TDD)

Financial Domain – Electronic Trading, Quantitative Research, Market Data Analytics, Real-time Analytics

PROJECTS

Real-Time Market Data Pipeline (HCL Technologies)

- Engineered a **Python**-based ETL pipeline to ingest and transform high-volume market data feeds, preparing them for loading into a **PostgreSQL** database for historical analysis.
- Developed **REST APIs** using **Flask** to serve aggregated market data snapshots to internal analytics applications, supporting approximately **2,000 requests per minute**.
- Implemented comprehensive logging and monitoring using **Splunk**, enabling rapid identification and resolution of data ingestion failures and performance bottlenecks.

Kdb+ Data Analytics Platform (HCL Technologies)

- Built **Kdb+/Q** scripts to perform time-series analysis on historical transaction data, identifying patterns related to trade execution anomalies for a financial services client.
- Designed and implemented **Q** functions for data aggregation and querying, reducing report generation time for key operational metrics by **40%**.
- Collaborated with business analysts to refine data requirements and ensure the delivered analytics met the needs of the trading operations team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)